Mini Project Guidelines: Big Data Analytics

Total marks: 25

Team size: 2 to 3 students **Duration:** 3 to 4 weeks

Synopsis Submission Link (**Deadline: 15th October 2026**):

V DSE - BDA Mini Project Synopsis Submission – Fill out form



Objective

Design and implement an end-to-end big data analytics solution on a **Hadoop cluster** using **Apache Spark**, from data ingestion to analysis and visualization.

Mandatory requirements

1. **Hadoop cluster:** a working HDFS and YARN setup with at least **One NameNode/ResourceManager and Two DataNodes/NodeManagers**. A multimode cluster, multi-VM setup (VirtualBox/VMware), Docker containers, or cloud instances are allowed. A single-node pseudo-distributed setup is not accepted.
2. **Data stored in HDFS:** the dataset must be loaded into HDFS and read from there by Spark.
3. **Apache Spark on YARN:** jobs must run in cluster or client mode on YARN, using PySpark or Scala. Show the job in the YARN/Spark UI.
4. **Spark component usage:** use at least **two** of: Spark SQL/DataFrames, Spark MLlib, Spark Structured Streaming, GraphX/GraphFrames.
5. **Dataset size:** at least **500MB** (or 10000+ records). Justify your choice if smaller.
6. **Original analysis:** a clear problem statement and meaningful insights, not just a word count.

Deliverables

1. **Project report** (15 to 25 pages) containing:
    - Title, team details, abstract
    - Problem statement and objectives

- Dataset description (source, size, schema, license)
- Architecture diagram and cluster configuration (nodes, RAM, cores, versions)
- Methodology and Spark code explanation
- Results with screenshots (HDFS UI, YARN UI, Spark UI, outputs)
- Conclusion, limitations, future work, references

2. **Source code:**
3. **Demo and viva:**

Evaluation scheme (20 marks)

Component	**Marks**	**What is assessed**
Hadoop cluster setup	4	Multi-node HDFS and YARN working, configuration
Spark implementation	5	Correct use of Spark on YARN, DataFrames/SQL/MLlib, code quality, data cleaning
Analysis and results	4	Quality of insights or model evaluation, visualization, interpretation
Report and documentation	7	Structure, clarity, references, README
Demo and viva	5	Live run, individual understanding and answers
Total	**25**	

Individual moderation: viva answers decide each member's share. A member who cannot explain the work may lose up to 50% of the marks.

Rules

- Each team must have a **unique topic and dataset combination**.
- Copied code or reports will receive zero marks. Cite any code or tutorial used. Plagiarism checks apply to reports.
- Do not use only the Spark local[*] mode. The YARN execution must be visible.
- Respect dataset licenses, and anonymize any personal data.

Synopsis Format

Length: 1 to 2 pages.

Submission: Word/PDF file in the given link (1 submission per team).
Formatting: A4, Times New Roman 12 pt, 1.5 line spacing, 1-inch margins.

Cover details

- Project title
- Course name and code, semester, department, Section
- Team members (name, roll number, register number)

Content :

No.	Section	What to write
1	**Title**	Specific and descriptive, e.g., "Customer Segmentation of E-commerce Transactions Using Spark MLlib on a Hadoop Cluster"
2	**Introduction and motivation**	Background of the problem and why big data tools are needed (5 to 8 lines)
3	**Problem statement**	One clear paragraph on what you will analyze or predict
4	**Objectives**	3 to 5 measurable points
5	**Dataset details**	Name, source link, size (GB or records), format, number of attributes, license, and a note on why it qualifies as big data
6	**References**	dataset page link